CO3 - AT1 - Case Study

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Case Study 1 — Employee Salaries

A company records the annual salaries (in ₹ lakhs) of 9 employees:
4, 5, 5, 6, 7, 8, 8, 9, 30
1. Find the mean, median, and mode.
2. Find the range.
3. Calculate the variance and standard deviation.
4. Identify the outlier.
5. Which measure of central tendency best represents the typical salary? Explain why

Solution:
Given Data
A company records the annual salaries (in ₹ lakhs) of **9 employees**:
4, 5, 5, 6, 7, 8, 8, 9, 30
The data is already arranged in ascending order.

1. Find the Mean, Median, and Mode

A) Mean (Average)

Formula

$$\text{Mean}=\frac{\sum X}{N}$$

Where:

- ΣX = Sum of all salaries
- N = Number of employees

Total salary

4 + 5 + 5 + 6 + 7 + 8 + 8 + 9 + 30 =82

Number of employees

N=9

Therefore,

$$\text{Mean} = \frac{82}{9}$$

$$= 9.11$$

Answer

Mean Salary = ₹9.11 lakhs

B) Median

Since there are 9 observations (odd number),

Median position = (9 + 1) / 2

= 5

Therefore, 5th value is 7.

Answer

Median = ₹7 lakhs

C) Mode

Frequency of each number

4	→	1
5	→	2
6	→	1
7	→	1
8	→	2
9	→	1
30	→	1

Both ₹5 lakhs and ₹8 lakhs occur twice.

Answer

Mode = ₹5 lakhs and ₹8 lakhs (Bimodal Distribution)

2. Find the Range

Formula

Range= Maximum Value − Minimum Value

Maximum salary - 30

Minimum salary - 4

Therefore,

30 - 4 = 26

Answer

Range = ₹26 lakhs

3. Calculate the Variance and Standard Deviation

Step 1: Mean

X = 9.11

Step 2: Compute Squared Deviations

Salary (X)	X − Mean	(X − Mean)²
4	-5.11	26.11
5	-4.11	16.89
5	-4.11	16.89
6	-3.11	9.68
7	-2.11	4.46

8	-1.11	1.23
8	-1.11	1.23
9	-0.11	0.01
30	20.89	436.37

Sum of squared deviations

= 512.88

Variance

Using the population formula:

$$\sigma^2 = \frac{\sum(X - \bar{X})^2}{N}$$

$$= \frac{512.88}{9}$$

= 56.99

Answer

Variance ≈ 56.99

Standard Deviation

Formula

$$\sigma = \sqrt{\sigma^2}$$

$$= \sqrt{56.99}$$

$= 7.55$

Answer

Standard Deviation ≈ ₹7.55 lakhs

4. Identify the Outlier

The salaries are:

4, 5, 5, 6, 7, 8, 8, 9, **30**

Most employees earn between ₹4 lakhs and ₹9 lakhs, whereas ₹30 lakhs is far above the rest.

Answer

Outlier = ₹30 lakhs

5. Which Measure of Central Tendency Best Represents the Typical Salary? Explain

The three measures of central tendency are:

- Mean = ₹9.11 lakhs
- Median = ₹7 lakhs
- Mode = ₹5 lakhs and ₹8 lakhs

Analysis

- The mean is affected by the outlier (₹30 lakhs), making it higher than the salaries earned by most employees.
- The mode only shows the most frequent salaries and does not summarize the entire dataset.
- The median remains unaffected by the outlier and represents the middle salary.

Answer

The median (₹7 lakhs) is the best measure of central tendency.

Reason

- It is not influenced by the extreme salary.
- It accurately represents the typical employee's annual salary.
- It provides a better measure of the center when the data contains an outlier.

Case Study 2 — Website Response Time

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

1. Calculate the mean and median.

2. Find the range.

3. Calculate the variance and standard deviation.

4. Determine whether the data is likely to be right-skewed or left-skewed.

5. Explain why the median may be more useful than the mean.

Solution:

Given Data

A data scientist records the response time (in milliseconds) of a website for 10 requests:

120, 125, 128, 130, 132, 135, 140, 145, 150, 300

The data is already arranged in ascending order.

1. Calculate the Mean and Median

A) Mean (Average)

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Where:

- ΣX = Sum of all salaries
- N = Number of employees

Sum of all response times:

120 + 125 + 128 + 130 + 132 + 135 + 140 + 145 + 150 + 300 =1505

Number of observations:

N=10

Therefore,

$$\text{Mean} = \frac{1505}{10}$$

$$= 150.5$$

Answer

Mean = 150.5 ms

B) Median

Since there are 10 observations (an even number), the median is the average of the 5th and 6th observations.

$$\text{Median} = \frac{132 + 135}{2}$$

$$= 133.5$$

Answer

Median = 133.5 ms

2. Find the Range

Formula

Range= Maximum Value − Minimum Value

Calculation

Maximum response time - 300 ms

Minimum response time - 120 ms

Therefore,

$$300 - 120 = 180$$

Answer

Range = 180 ms

3. Calculate the Variance and Standard Deviation

Step 1: Mean

$$\bar{X} = 150.5$$

Step 2: Calculate the Squared Deviations

Response Time (X)	X − Mean	(X − Mean)²
120	−30.5	930.25
125	−25.5	650.25
128	−22.5	506.25
130	−20.5	420.25
132	−18.5	342.25
135	−15.5	240.25

140	−10.5	110.25
145	−5.5	30.25
150	−0.5	0.25
300	149.5	22350.25

Sum of squared deviations:

=25580.50

Variance

$$\sigma^2 = \frac{\sum(X-\bar{X})^2}{N}$$

$$= \frac{25580.50}{10}$$

$$= 2558.05$$

Answer

Variance ≈ 2558.05 ms²

Standard Deviation

Formula

$$\sigma = \sqrt{\sigma^2}$$

$$= \sqrt{2558.05}$$

$$= 50.58$$

Answer

Standard Deviation ≈ 50.58 ms

4. Determine Whether the Data is Right-Skewed or Left-Skewed

Observation

Most response times lie between 120 ms and 150 ms, while 300 ms is much larger than the rest.

Also,

- Mean = 150.5 ms
- Median = 133.5 ms

Since the mean is greater than the median, the data is positively skewed (right-skewed).

Answer

The data is right-skewed (positively skewed).

5. Explain Why the Median May Be More Useful Than the Mean

Mean

- Uses every observation in the dataset.
- Is affected by extremely high or low values (outliers).

Median

- Represents the middle observation.
- Is resistant to outliers.
- Better reflects the typical response time when extreme values exist.

Comparison

Measure	Value

Mean	150.5 ms
Median	133.5 ms

The single response time of 300 ms increases the mean by a significant amount. However, the median remains close to the response times experienced during normal website operation.

Answer

The median (133.5 ms) is more useful because it is not affected by the unusually slow response time of 300 ms. It provides a more accurate representation of the website's typical response time and is therefore a better measure of central tendency for this dataset.

Case Study 3 — Student Performance

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

1. Find Q1, Q2 and Q3.

2. Calculate the IQR.

3. Use the 1.5 × IQR rule to identify any outliers.

4. Find the mean and median.

5. Explain how the outlier affects the mean.

Solution:

Given Data

The marks of 12 students in a Data Science test are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

The data is already arranged in ascending order.

1. Find Q1, Q2, and Q3

Definition

Quartiles divide a dataset into **four equal parts**.

- **Q1 (First Quartile)** = 25% of the data lies below this value.
- **Q2 (Second Quartile)** = Median (50% of the data lies below this value).
- **Q3 (Third Quartile)** = 75% of the data lies below this value.

Since there are **12 observations (even number)**, the data is divided into two equal halves.

Lower Half

42, 45, 48, 50, 52, 52

Upper Half

55, 58, 60, 62, 65, 95

A) First Quartile (Q1)

Q1 is the median of the lower half.

Middle values:

48 and 50

$$Q1 = \frac{48 + 50}{2}$$

$$= 49$$

Answer

Q1 = 49

B) Second Quartile (Q2 / Median)

The median is the aver**age of the 6th and 7th observations.**

6th observation = 52

7th observation = 55

$$Q2 = \frac{52 + 55}{2}$$

$$= 53.5$$

Answer

Q2 (Median) = 53.5

C) Third Quartile (Q3)

Q3 is the median of the upper half.

Middle values:

60 and 62

$$Q3 = \frac{60 + 62}{2}$$

$$= 61$$

Answer

Q3 = 61

2. Calculate the Interquartile Range (IQR)

Formula

$$IQR = Q3 - Q1$$

Calculation

$$IQR = 61 - 49 = 12$$

Answer

IQR = 12

3. Use the 1.5 × IQR Rule to Identify Any Outliers

Step 1: Calculate 1.5 × IQR

$$1.5 \times 12 = 18$$

Step 2: Find the Lower Limit

$$Q1 - 1.5(IQR) = 49 - 18 = 31$$

Step 3: Find the Upper Limit

$$Q3 + 1.5(IQR) = 61 + 18 = 79$$

Therefore,

- Lower Limit = **31**
- Upper Limit = **79**

Any value below 31 or above 79 is considered an outlier.

Observation

The marks are:

42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Only 95 is greater than 79.

Answer

Outlier = 95

4. Find the Mean and Median

A) Mean

Formula

$$\text{Mean} = \frac{\sum X}{N}$$

Total marks:

$$42 + 45 + 48 + 50 + 52 + 52 + 55 + 58 + 60 + 62 + 65 + 95 = 684$$

Number of students:

N=12

Mean = $\frac{684}{12}$ = 57

Answer

Mean = 57

B) Median

The median has already been calculated as:

53.5

Answer

Median = 53.5

5. Explain How the Outlier Affects the Mean

In this dataset, the score 95 is much higher than the remaining marks.

Effect on the Mean

The mean uses every value in the dataset. Since **95** is significantly larger than the other scores, it increases the total sum of marks and raises the average.

- **Mean = 57**
- **Median = 53.5**

The mean is higher than the median because of the influence of the outlier.

Effect on the Median

The median depends only on the middle observations and is not affected by extreme values. Even though one student scored 95, the median remains 53.5, which better represents the typical student's performance.

Answer

The outlier **95** increases the mean, making the average appear higher than the performance of most students. The median remains unchanged and therefore provides a more accurate measure of the typical student's score.

Case Study 4 — E-Commerce Orders

An online store records the number of orders received over 10 days:
18, 20, 22, 22, 24, 25, 26, 28, 30, 35
1. Calculate the mean, median, and mode.
2. Find the variance and standard deviation.
3. Calculate the coefficient of variation if the mean and standard deviation are known.
4. What does the standard deviation tell the company?
5. Would you describe the order data as highly or moderately variable?

Solution:

Given Data

An online store records the number of orders received over 10 days:

18, 20, 22, 22, 24, 25, 26, 28, 30, 35

The data is already arranged in ascending order.

1. Calculate the Mean, Median, and Mode

A) Mean (Average)

Definition

The mean is the average number of orders received per day. It is calculated by dividing the total number of orders by the number of days.

Formula

$$\text{Mean}= \frac{\sum X}{N}$$

Total number of orders:

$$18+ 20 + 22 + 22 + 24 + 25 + 26 + 28 + 30 + 35 = 250$$

Number of days:

N=10

Mean = $\frac{250}{10}$ = 25

Median

$\frac{24 + 25}{2}$ = 24.5

Answer: Median = 24.5 orders

Mode

22 occurs twice.

Answer: Mode = 22 orders

2. Variance and Standard Deviation

Mean = 25

Variance:

$$\sigma^2 = \frac{\sum(X - \bar{X})^2}{N}$$

$$= \frac{228}{10}$$

= 22.8

Standard Deviation:

$$\sigma = \sqrt{\sigma^2}$$

$$= \sqrt{22.8}$$

Answer:

- Variance = 22.8
- Standard Deviation = 4.77 orders

3. Coefficient of Variation (CV)

Formula:

$$CV = \frac{\sigma}{Mean} X\ 100$$

$$= \frac{4.77}{25} X\ 100$$

= 19.08%

Answer: CV = 19.08%

4. What does the Standard Deviation tell?

The standard deviation (4.77) shows that the daily orders vary by about **5 orders** from the average. It helps the company plan inventory, staff, and deliveries.

5. Is the data highly or moderately variable?

Since the **CV is 19.08%**, the data has **moderate variability**. The daily orders are fairly consistent without large fluctuations.

Case Study 5 — Comparing Two Data Science Models

Two machine-learning models have the following prediction errors:

Model A: 2, 3, 3, 4, 4, 5, 5, 6

Model B: 1, 1, 2, 4, 5, 7, 8, 9

1. Calculate the mean error for both models.

2. Calculate the variance for both models.

3. Calculate the standard deviation for both.

4. Which model has more consistent predictions?

5. Can the model with the lower mean error necessarily be considered better? Explain.

Solution:

Given Data
Model A: 2,3,3,4,4,5,5,6
Model B: 1,1,2,4,5,7,8,9

1. Calculate the Mean Error
Formula:

Mean = $\Sigma X / N$

Model A:

$\Sigma X = 2+3+3+4+4+5+5+6 = 32$
$N=8$
Mean = $32/8 = 4$

Model B:

$\Sigma X = 1+1+2+4+5+7+8+9 = 37$
$N=8$
Mean = $37/8 = 4.625$

2. Calculate the Variance
Formula:

Variance = $\Sigma(X\text{-Mean})^2 / N$

Model A (Mean = 4)

X	X-4	(X-4)^2
2	-2	4
3	-1	1
3	-1	1
4	0	0
4	0	0
5	1	1

5	1	1
6	2	4

$\Sigma(X\text{-Mean})^2 = 12$
Variance = 12/8 = 1.50

Model B (Mean = 4.625)

X	X-4.625	(X-4.625)^2
1	-3.625	13.141
1	-3.625	13.141
2	-2.625	6.891
4	-0.625	0.391
5	0.375	0.141
7	2.375	5.641
8	3.375	11.391
9	4.375	19.141

$\Sigma(X\text{-Mean})^2 = 69.875$
Variance = 69.875/8 = 8.73

3. Calculate the Standard Deviation
Formula:
Standard Deviation = $\sqrt{\text{Variance}}$

Model A: $\sqrt{1.50} = 1.22$
Model B: $\sqrt{8.73} = 2.96$

4. Which Model Has More Consistent Predictions?
Comparison:
Mean Error: A=4, B=4.625
Variance: A=1.50, B=8.73
Standard Deviation: A=1.22, B=2.96

Model A has the lower variance and standard deviation, so it is more consistent.

5. Can the Model with the Lower Mean Error Necessarily Be Considered Better?
No. Mean error alone is not sufficient to judge a model. A good model should have:
- Low mean error
- Low variance
- Low standard deviation
- Consistent predictions

From the calculations:
Model A: Mean=4, SD=1.22
Model B: Mean=4.625, SD=2.96
Hence, Model A is better because it has both lower average error and more consistent predictions.